DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

TLE	2. IDENTIFICATION NUMBER
SEMICONDUCTOR PROCESS SPECIFICATION	DI- QCIC-80924

3. DESCRIPTION/PURPOSE

3.1 The Semiconductor Process Specification describes the electrical parameters/characteristics and the topological design rules (artwork layout) required for the fabrication of semiconductor integrated circuits. It includes information on the derivation of these parameters, and the testing requirements to verify accuracy.

(Continued on Page 2)

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
900105	G/T213		

7 APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

7.2 This DID is used on systems/equipment acquisitions where semi-conductor integrated circuits are designed and fabricated.

7.3 This DID supersedes DI-E-5444.

8 APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b AMSC NUMBER
		S4865

PREPARATION INSTRUCTIONS

10.1 Format requirements. The data shall be prepared in contractor format on 8 1/2" X 11" (metric size A4) bond paper. Drawings and illustration fold-outs may be used, but must not exceed the 8 1/2" x 11" limits when folded. Photo-reduction of over-sized pages is preferred, provided such reductions are easily readable and reproducible.

10.2 Content. The report shall contain the following information:

10.2.1 Cover/title page. The following information shall appear on the cover and title page:

a. Title.
b. Sequence number.
c. Contract number.
d. Contractor name and address.
e. Processing facility name and address.
f. Report name.

10.3 Body of Report.

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DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Block 3, Description/Purpose (Continued)

The data is used as a procurement specification to assure that equivalent electrical circu performance can be obtained from suppliers.

Block 10, Preparation Instructions (Continued)

10.3.1 Electrical parameters/characteristics. For each of the design parameters listed in Table 1, the following shall be specified:

10.3.1.1 Parameter value/characteristics in the form of histograms, distributions, minimum/maximum, or typical values.

10.3.1.2 The sample size upon which the values were based, where applicable.

10.3.1.3 Information on parametric circuits used to determine and quantify the parameters.

10.3.1.4 The test set-up procedures, and conditions under which the parameter values were determine.

TABLE I. Electrical Parameters

Parameter	Parameter
1. Starting material	9. Junction breakdowns
2. Diffusion resistivity	10. Doping concentrations
3. Mobility	11. Body effects
4. Oxide thickness	12. Punch-through voltages
5. Diffusion depth	13. Metal resistances
6. Gate and field thresholds	14. Delta threshold limits under stress biasing
7. Gain factors	15. Other as applicable
8. Capacitances	

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Block 10, Preparation Instructions (Continued)

.2 Topographical design rules. This section describes the Topographical (artwork layout) design rules to be followed in the layout of the fabrication process. For each of the applicable areas listed below, a pictorial representation of the layout rule shall be included reflecting associated minimum, maximum, or typical values sufficient to permit layout or modification of the circuit layout.

a. Line widths.

b. Line spacing.

c. Transistor device geometric.

d. Contact design for level-to-level interconnect.

e. Diffusion widths.

f. Difussion-difussion spacing/overlap.

g. Bonding pad sizes and spacing to scribe line and circuit areas.

h. Scribe area description.

i. Alignment marks.

j. Standard test devices.

k. Numerical mask and artwork level identification, including:

1. Level nomenclature
2. Fiduciary markings
3. Coloring/crosshatching used to distinguish levels
4. Polarity of each mask level

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